Addis Ababa Science & Technology University
Department of Physics

General Physics (phys1001)

Chapter - 3
Fluid Mechanics

FLUID MECHANICS

- Fluid mechanics: is the branch of physics concerned with the mechanics of fluids in motion (**fluid dynamics**) or at rest (**fluid statics**) and the forces on them.

- Fluid: a fluid is a substance that continually deforms (flows) under an applied shear stress, or external force.

- A Fluid is a collection of molecules that are randomly arranged and held together by weak cohesive forces and by forces exerted by the walls of a container

- Fluids are a phase of matter and include liquids and gases.

Properties of Bulk Matter

Elastic materials: materials that regain their original shape and size when the deforming force is removed.

Elastic deformation: a reversible deformation by a force applied within the elastic limit.

Beyond elastic limit, a force applied on an object causes permanent and irreversible deformation called plastic deformation.

Plastics materials : do not regain their original shape and size when the deforming force is removed.

Examples of elastic materials

These materials all change shape when a reasonable force is applied, but will return to their original shape and size when the force is removed

The elastic properties of solid materials are described in terms of stress and strain.

Stress: is the force per unit area that is causing some deformation on an object.

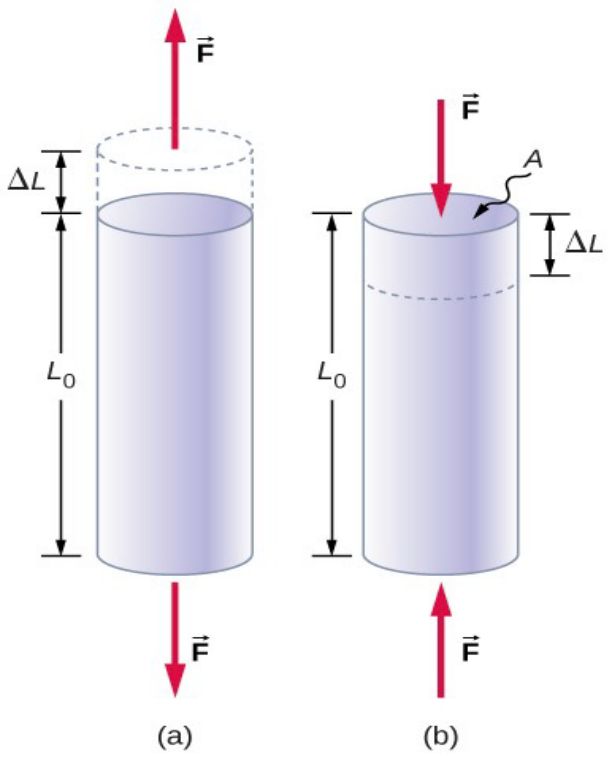
Strain : measures the amount of deformation by the applied stress and defined as the change in configuration of a body divided by its initial configuration.

Stress

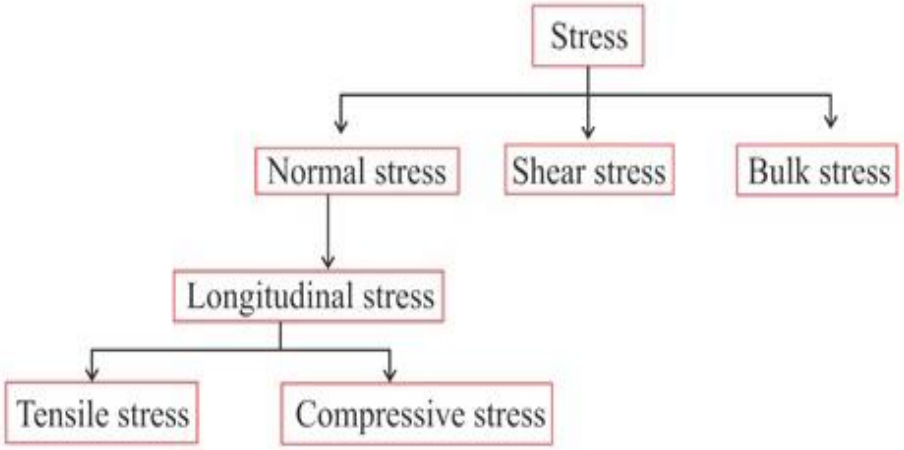
$$Stress = \frac{Force\ (or\ load)}{Area} \qquad \sigma = \frac{F}{A}$$

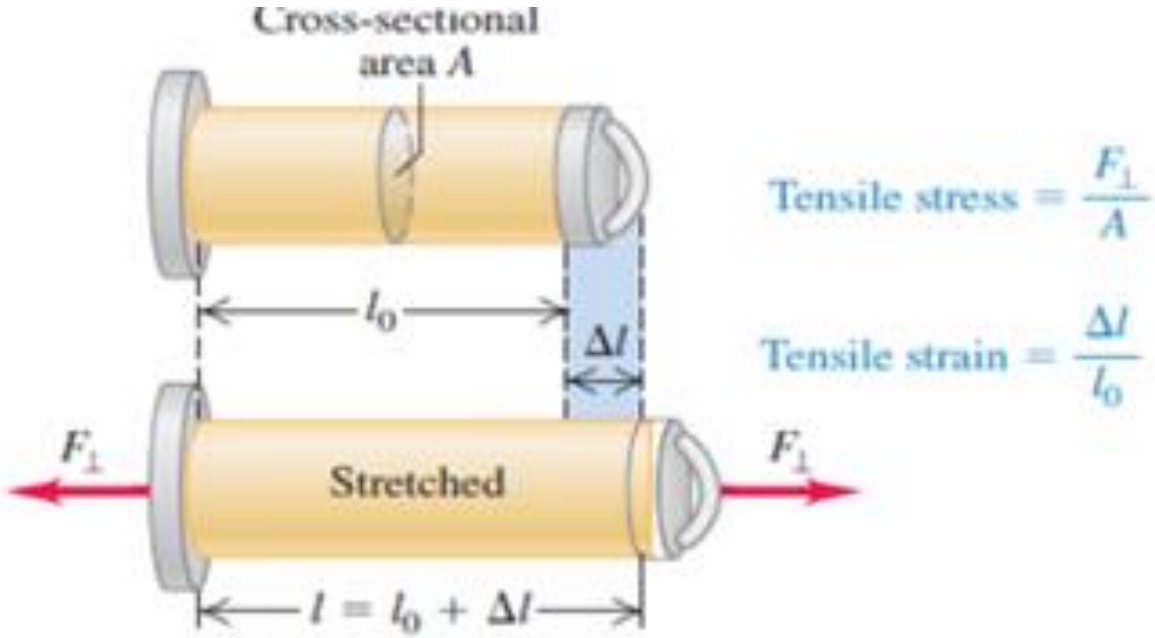
Strain

$$Strain = \frac{Length\ of\ Stretch}{Original\ Length} \qquad \varepsilon = \frac{\delta}{L}$$



1. Tensile Strain and Tensile stress

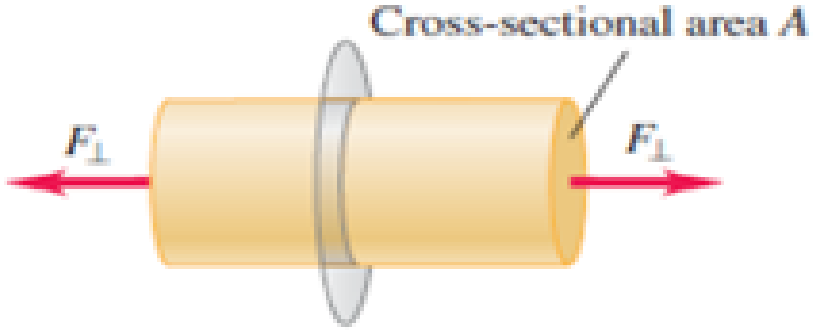




(b) Tensile strain

$$\text{Tensile stress} = \frac{F_\perp}{A}$$

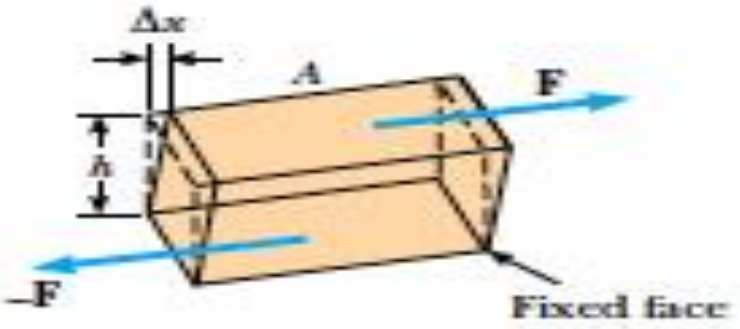
$$\text{Tensile strain} = \frac{\Delta l}{l_0}$$



(a) A bar in tension

2.Shear Strain

Another type of deformation occurs when an object is subjected to a force parallel to one of its faces while the opposite face is held fixed by another force



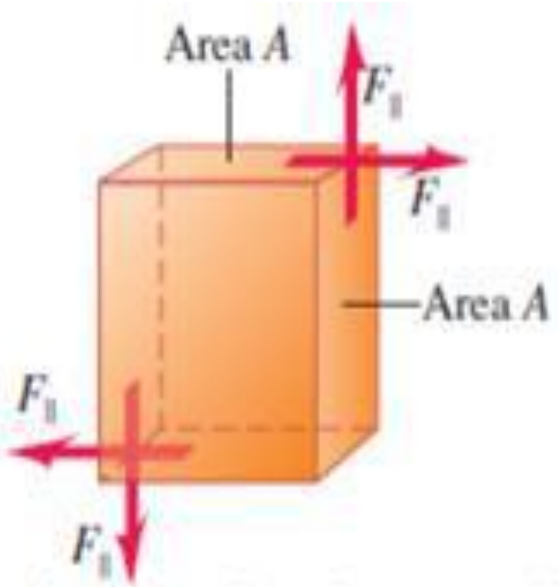
(a)

$$Shear\ strain = \frac{x}{h} = \tan\phi.$$

(b)

(a)A shear deformation in which a rectangular block is distorted by two forces of equal magnitude but opposite directions applied to two parallel faces. (b) A book under shear stress.

3. Shear stress

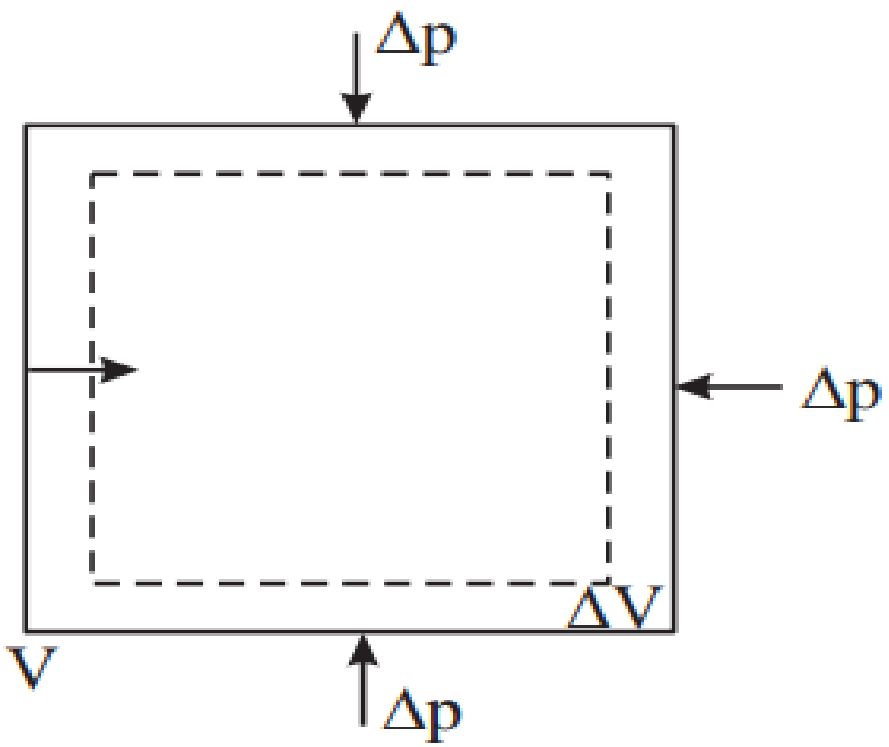


A block subjected to shear stress from two directions

(a) Shear stress.

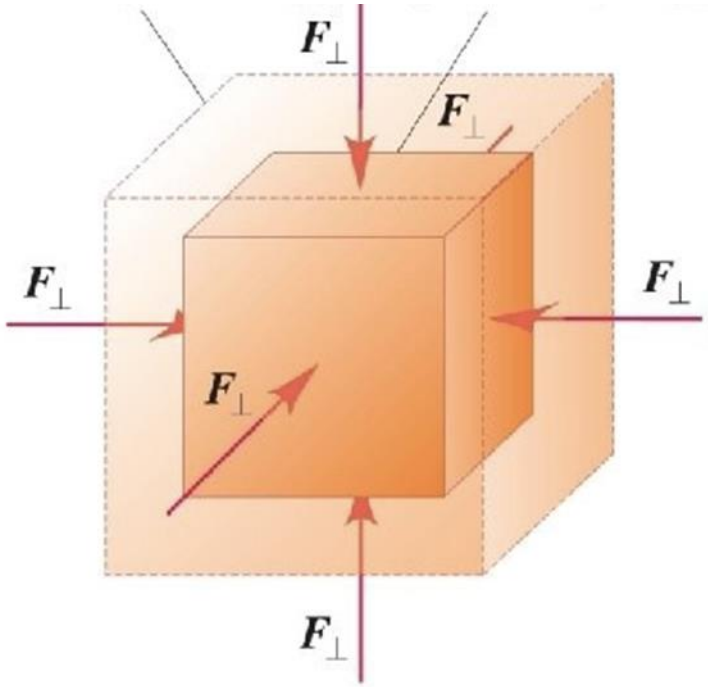
$$Shear\ stress = \frac{\Delta F}{A}.$$

Volume Stress



$$Volume\ Stress = change\ in\ Pressure = \Delta P$$

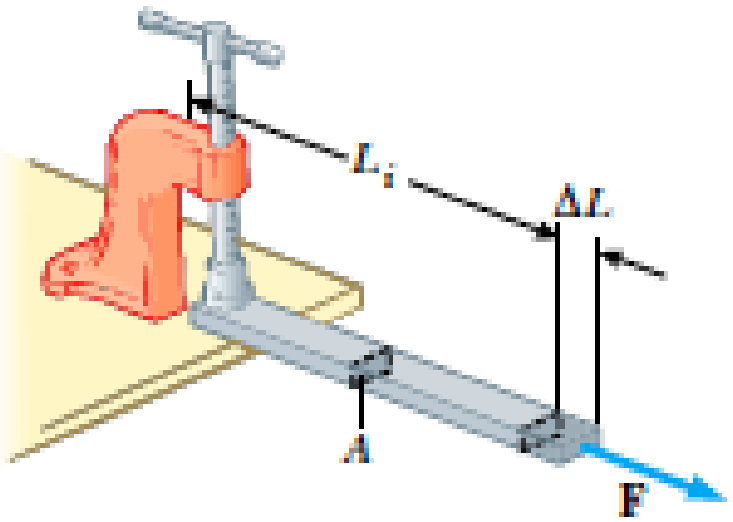
Volume Strain



$$\text{Volume strain} = \frac{\Delta V}{V_0}$$

1. **Young's modulus-Elasticity in Length**

It measures the resistance of a solid to a change in its length

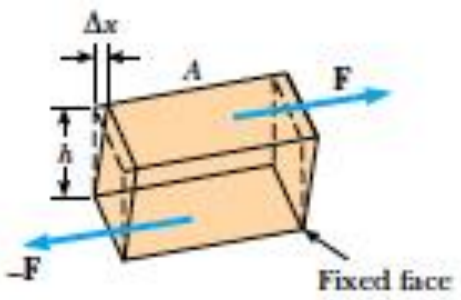


$$Y = \frac{\text{tensile stress}}{\text{tensile strain}} = \frac{F/A}{\Delta L/L_i}$$

2. Shear modulus-Elasticity of Shape

- It measures the resistance to motion of the planes within a solid parallel to each other

- Another type of deformation occurs when an object is subjected to a force parallel to one of its faces while the opposite face is held fixed by another force



(a)

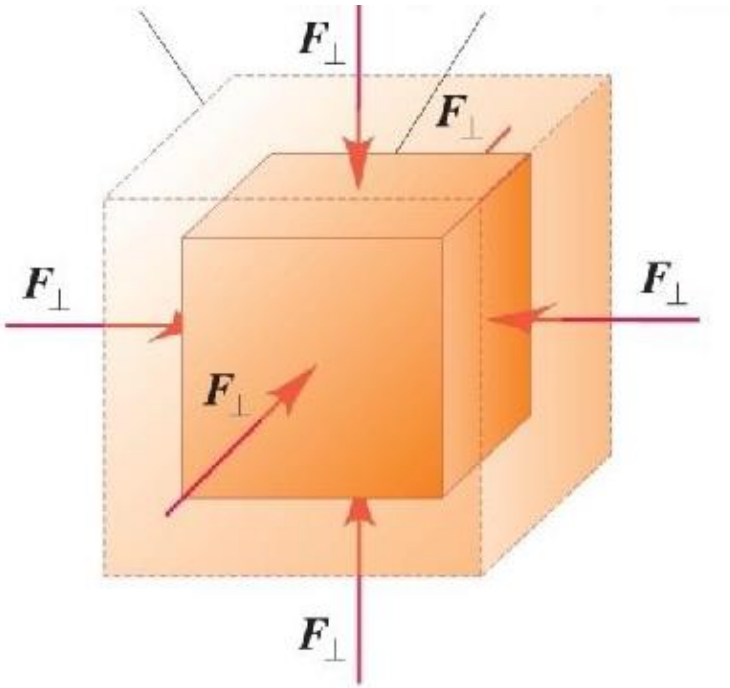
(b)

$$S = \frac{\text{shear stress}}{\text{shear strain}} = \frac{F/A}{\Delta x/h}$$

3. **Bulk modulus: Elastic in volume**

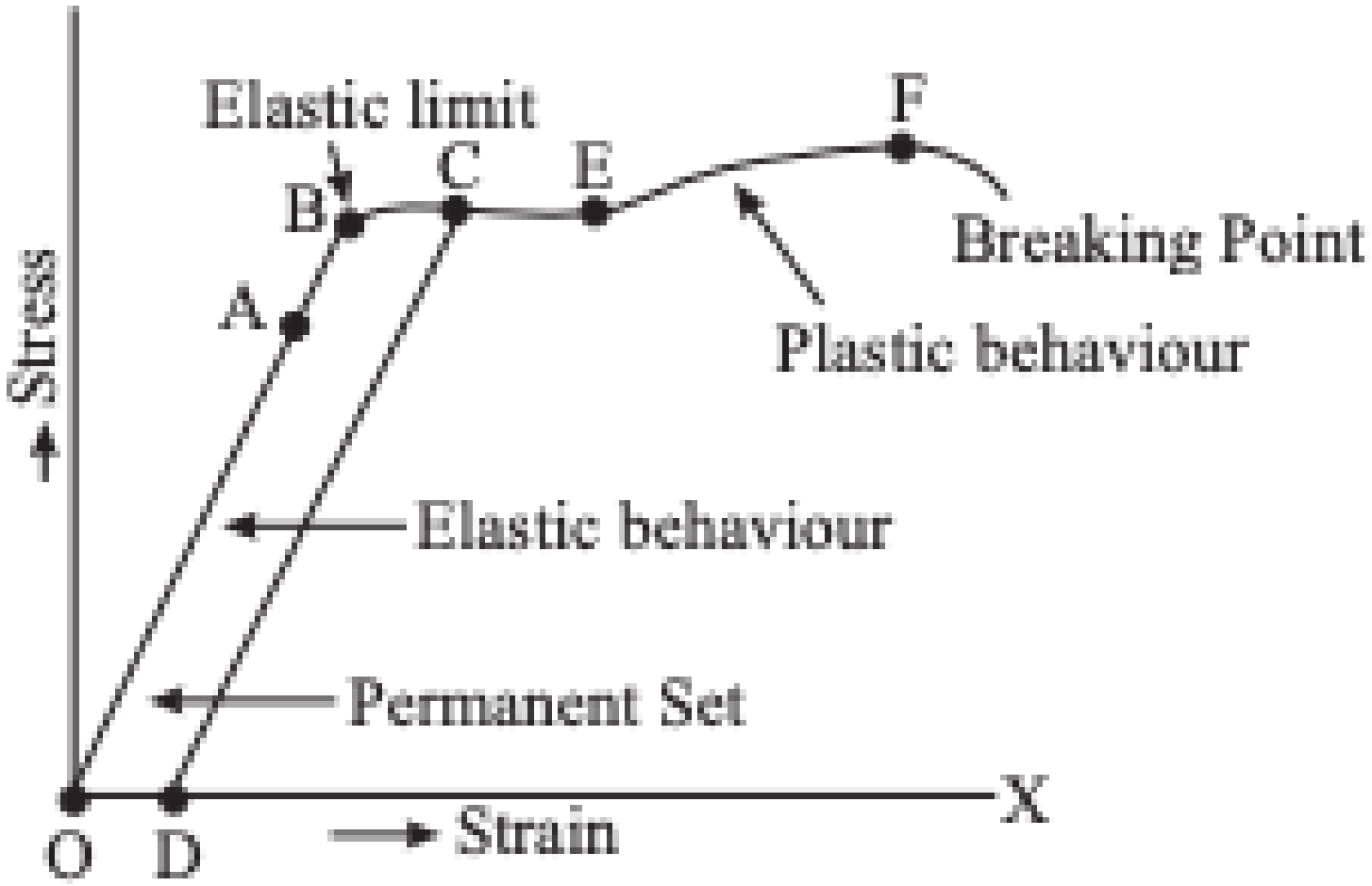
- It measures the resistance of solids or liquids to changes in their volume

- Bulk modulus characterizes the response of an object to changes in a force of uniform magnitude applied perpendicularly over the entire surface of the object,

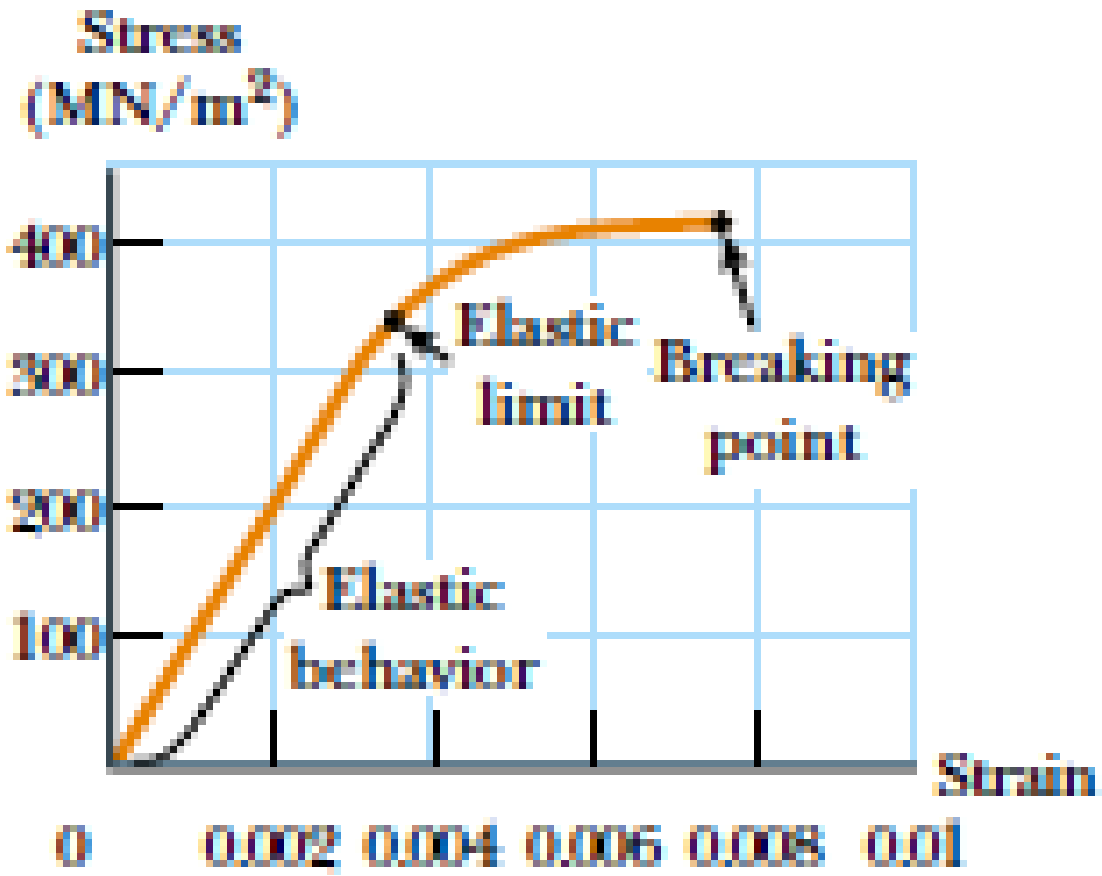


$$Bulk\ Modulus = \frac{Volume\ Stress}{Volume\ Strain} = \frac{V_0 \Delta P}{\Delta V}$$

Stress-strain Curve for a Metallic Wire



STRESS STRAIN CURVE



Typical Values for Elastic Moduli

Substance	Young's Modulus (N/m^2)	Shear Modulus (N/m^2)	Bulk Modulus (N/m^2)
Tungsten	35×10^{10}	14×10^{10}	20×10^{10}
Steel	20×10^{10}	8.4×10^{10}	6×10^{10}
Copper	11×10^{10}	4.2×10^{10}	14×10^{10}
Brass	9.1×10^{10}	3.5×10^{10}	6.1×10^{10}
Aluminum	7.0×10^{10}	2.5×10^{10}	7.0×10^{10}
Glass	$6.5–7.8 \times 10^{10}$	$2.6–3.2 \times 10^{10}$	$5.0–5.5 \times 10^{10}$
Quartz	5.6×10^{10}	2.6×10^{10}	2.7×10^{10}
Water	—	—	0.21×10^{10}
Mercury	—	—	2.8×10^{10}

1. A load of 100 kg is suspended by a wire of length 1.0 m and cross sectional area 0.10 cm^2. The wire is stretched by 0.20 cm. Calculate the (i) tensile stress, and (ii) strain in the wire. Given, g = 9.80 ms^{-2}.

Solution :

$$\text{(i) Tensile stress} = \frac{F}{A} = \frac{Mg}{A}$$

$$= \frac{(100\ \text{kg})\ (9.80\ \text{ms}^{-2})}{0.10 \times 10^{-4}\ \text{m}^2}$$

$$= 9.8 \times 10^7\ \text{Nm}^{-2}$$

$$\text{(ii) Tensile strain} = \frac{\Delta \ell}{\ell} = \frac{0.20 \times 10^{-2}\ \text{m}}{1.0\ \text{m}}$$

2. Calculate the maximum length of a steel wire that can be suspended and Fluids without breaking under its own weight, if its breaking stress = 4.0 x 108 Nm^{-2}, density = 7.9 × 103 kg m–3 and g = 9.80 ms^{-2}

Solution : The weight of the wire $W = A\ell\rho g$, where, A is area of cross section of the wire, ℓ is the maximum length and ρ is the density of the wire. Therefore, the breaking stress developed in the wire due to its own weight $\frac{W}{A} = \rho l g$. We are told that

breaking stress is 4.0 x 10^8 Nm^{-2}. Hence

$$\ell = \frac{4.0\times10^8\ \mathrm{Nm^{-2}}}{(7.9\times10^3\ \mathrm{kg\ m^{-3}})\,(9.8\ \mathrm{ms^{-2}})}$$

$$= 0.05 \times 10^5\ \mathrm{m}$$

$$= 5 \times 10^3\ \mathrm{m} = 5\ \mathrm{km}.$$

Now it is time to take a break and check your understanding

Density and Pressure in Static Fluids

- The quantity which is useful to compare the masses of different substances of equal volumes is the **density.**

- Density (ρ) : is the quantity of mass (m) per unit volume (V) of a body

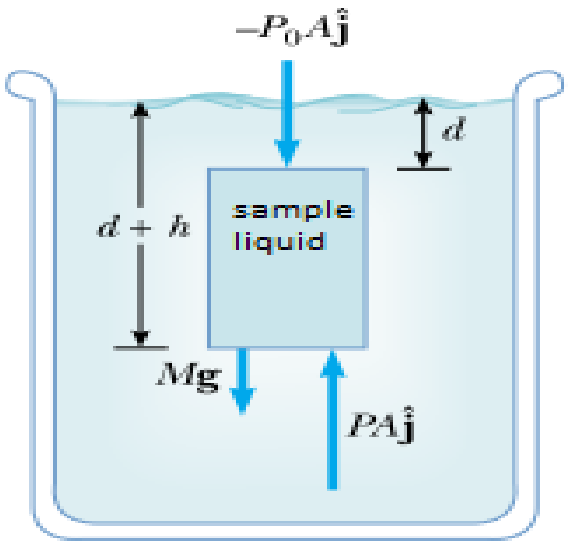
$$\rho = \frac{m}{V}$$

- Two objects made of the same material have the same density even though they may have different masses and different volumes.

- Specific gravity (SG): is the ratio of the density of the substance to the density of another substance which is taken as a standard(water.)

Pressure is the ratio of the force acting perpendicular to surface to the surface area (A) on which the force acts.

$$P = \frac{F_{\perp}}{A}$$



Variation of Pressure with Depth

Consider a liquid of density ρ at rest as shown in Fig. contained within an imaginary cylinder of cross-sectional area **A** extending from depth **d** to depth **d + h**.

$$\sum \mathbf{F} = PA\hat{\mathbf{j}} - P_0A\hat{\mathbf{j}} - M\mathbf{g}\hat{\mathbf{j}} = 0$$

$$PA - P_0A - \rho Ahg = 0$$

$$PA - P_0A = \rho Ahg$$

$$P = P_0 + \rho gh$$

$$P_0 = 1.00 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$$

Static fluid pressure does not depend on the shape, total mass, or surface area of the liquid.

$$\text{Pressure} = \frac{\text{weight}}{\text{area}} = \frac{mg}{A} = \frac{\rho Vg}{A} = \rho gh$$

- The pressure produced by the column of fluid of height **h** and density ρ is given by:

$$P_{fluid} = \rho gh$$

(this is, the pressure P at a depth h below a point in the liquid)

Note that:

- All points at same level in a fluid have same pressure.
- Fluid pressure increases with increase in the depth of the fluid.
- Fluid pressure does not depend on the shape of the container.

Example :-

1. A girl of mass 50kg is wearing high-heeled shoes with heels of total area $5cm^2$, what pressure will she exert when all her weight rests on the heel? $\left(Take\ g = 9.8\frac{m}{s^2}\right)$

Solution ; the girl exerts pressure due to her weight , her weight is $w = mg\ = 50kgx9.8\ ms^{-2}$ =490 N

The pressure she exerts P $= \frac{F}{A}$, with $F = 490N, and\ A = 5cm^2$ = $5x10^{-4}m^2$,then the pressure

$$P = \frac{490N}{5x10^{-4}m^2} = 9.8x10^5\frac{N}{m^2} \quad \therefore \quad P = 9.8x10^5\ pa$$

2. calculate the pressure at a depth of 0.6m in kerosene whose density is $800\frac{kg}{m^3}$, what would be the pressure at the same depth if the liquid were water?

Solution:

$\rho_{keros} = 800\frac{kg}{m^3}$, $h = 0.6m$, $g = 9.8\frac{m}{s^2}$, the pressure is calculated using $P = \rho hg$

$$P = 800\frac{kg}{m^3}x9.8\frac{m}{s^2}x0.6m = 4704\ \frac{N}{m^2}$$

$$\therefore \quad P = 4704\ pa$$

If the liquid were water, $\rho_{water} = 1000\frac{kg}{m^3}$ and the pressure would be

$$P = \rho_{water}gh = 1000\frac{kg}{m^3}x9.8\frac{m}{s^2}x0.6m = 5880\ \frac{N}{m^2}$$

$$\therefore \quad P = 5880pa$$

3. how much length of mercury will exert pressure equal to that of 34m of water? Given density of mercury $\rho_{Hg} = 13600\frac{kg}{m^3}$ and density of water $\rho_{HO_2} = 1000\frac{kg}{m^3}$

Solution : let h_1 be height of mercury and h_2 be height of water column

ρ_1 be density of mercury and ρ_2 be density of water

We are given that $\rho_1 h_1 g = \rho_2 h_2 g$ where g is constant

$$\Longrightarrow \quad \rho_1 h_1 = \rho_2 h_2$$

$$\Longrightarrow \quad h_1 = \frac{\rho_2}{\rho_1} h_2 \quad \Longrightarrow \quad h_1 = \frac{1000kgm^{-3}}{13600kgm^{-3}} x34m = \frac{34x10^3}{136x10^2} m$$

$$\Longrightarrow \quad h_1 = \frac{340}{136} m = 2.5\text{m}$$

4. A container has a 10cm thick layer of oil on 28cm of water. What is the total pressure of the fluids at the bottom of the container where the density of oil is 0.85 $\frac{gm}{cm^3}$?

Solution: the pressure at the bottom is the sum of pressure exerted by the oil and water

$$P_{bottom} = P_o + P_w$$

$$= \rho_o h_o g + \rho_w h_w g$$

$$= g(\rho_o h_o + \rho_w h_w)$$

$$= 9.8\frac{m}{s^2}(850kgm^{-3}x0.1m + 1000kgm^{-3}x0.28m)$$

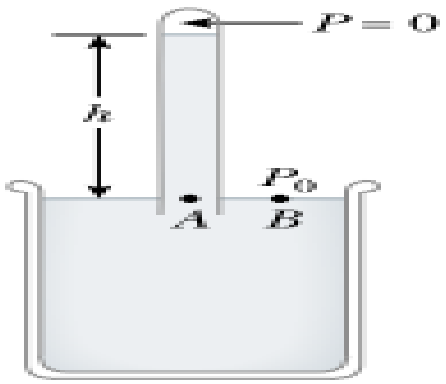
$$= 9.8\frac{m}{s^2}(85kgm^{-2} + 280kgm^{-2})$$

$$= 9.8\frac{m}{s^2}(365kgm^{-2})$$

$$= 3577\frac{N}{m^2} = 3577\,pa$$

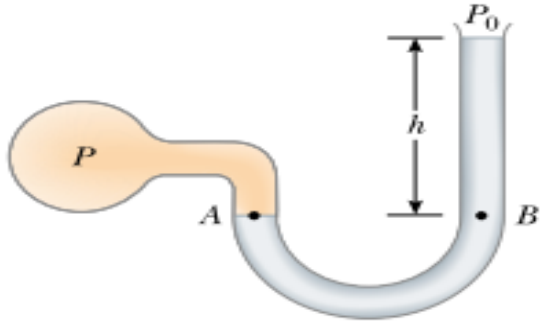
Pressure Measurements

1. A mercury Barometer : used to measure atmospheric pressure



$$P_0 = \rho_{Hg} g h \longrightarrow h = \frac{P_0}{\rho_{Hg} g} = \frac{1.013 \times 10^5 \text{ Pa}}{(13.6 \times 10^3 \text{ kg/m}^3)(9.80 \text{ m/s}^2)} = 0.760 \text{ m}$$

2. Open-tube Manometer: used to measure the pressure of a gas contained in a vessel



$$P = P_0 + \rho g h.$$

The pressure P is called the **absolute pressure**.

- The difference **P - Po** is called the Gauge pressure.

Atmospheric Pressure: is the pressure due to the weight of the atmosphere exerted on the surface of the Earth. Atmospheric pressure decreases with increase in altitude as a result of decrease in the density of the air.

Gauge pressure: is the difference in pressure between a system and the surrounding atmosphere.

$$P_{guage} = P_{system} - P_{atmosphere}$$

Absolute Pressure:

Also called the total pressure, is the sum of gauge pressure and atmospheric pressure:

$$P_{absolute} = P_{gauge} + P_{atmospheric}$$

In most cases the absolute pressure in fluids cannot be negative. Fluids push rather than pull, so the smallest absolute pressure is zero. (A negative absolute pressure is a pull.) Thus the smallest possible gauge pressure is $P_{gauge} = -P_{atmosphere}$

Example

1. A submerged wreck is located 18.3 m beneath the surface of the ocean. Determine a) gauge pressure and b) absolute pressure .

   Note: the density of sea water = 1025 kg/m3

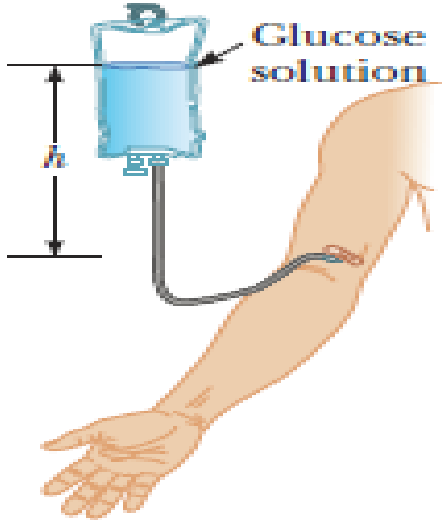
a) $P_{gauge} = \rho g h = (1025\ kg/m^3)(9.80\ m/s^2)(18.3\ m) = 1.83 \times \frac{10^5 N}{m^2} = 182\ kPa.$

b) $P_{abs} = P_{gau} + P_{atm} = (1.83 \times 10^5 + 1.013 \times 10^5)\ N/m^2 = 2.84 \times 10^5\ N/m^2 = 2.84 kPa\ .$

2. A collapsible plastic bag (see figure) contains a glucose solution. If the average gauge pressure in the vein is $1.33 \times 10_3$ Pa, what must be the minimum height *h* of the bag to infuse glucose into the vein? Assume the specific gravity of the solution is 1.02.

$$P_{guage} = \rho g h$$

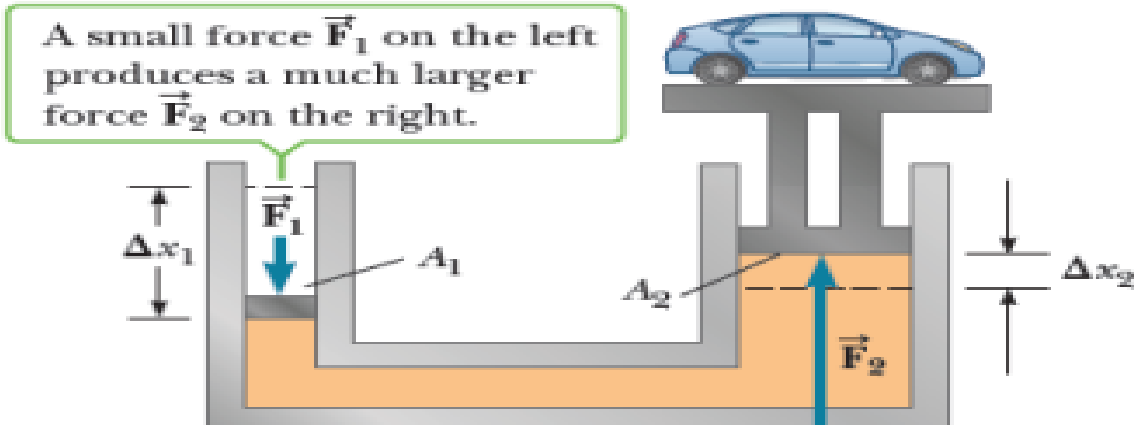
$$h = P_{guage} / \rho g$$



Pascal's Principle- states that pressure applied to a confined fluid in a container is transmitted equally to all regions of the fluid and to the walls of the container.

$$P_1 = P_2$$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$



- **An important application of Pascal's principle is the hydraulic press**
- A force of magnitude F_1 is applied to a small piston of surface area A_1. The pressure is transmitted through an incompressible liquid to a large piston of surface area A_2 because the pressure must be the same on both sides.

$$P_1 = P_2 \Rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow \frac{F_2}{F_1} = \frac{A_2}{A_1}$$

The quantity $\frac{F_2}{F_1}$ is called the mechanical advantage of the hydraulic press which is equal to the ratio of the areas. Therefore, the force F_2 is greater than the force F_1 by a factor of $\frac{A_2}{A_1}$

- Hydraulic press is a force multiplier

1. The small piston of a hydraulic lift has an area of $0.20\ m^2$. A car weighing $1.2 \times 10^4 N$ sits on a rack mounted on the large piston. The large piston has an area of $0.90\ m^2$. How large force must be applied to the small piston to support the car?

Given:
$A_1 = 0.20\ m^2$

$A_2 = 0.90\ m^2$
$F_2 = 1.2 \times 10^4 N$

Unknown:

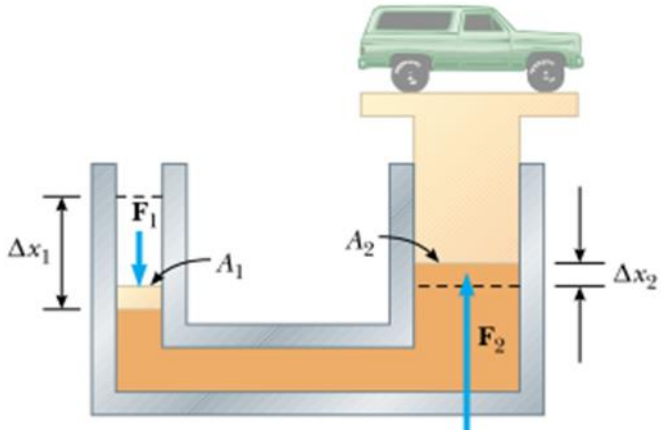
$F_1 = ?$

Solution:

$$F_1/A_1 = F_2/A_2 \Rightarrow F_1 = F_2\left(A_1/A_2\right)$$

$$= (1.2 \times 10^4 N)\left(\frac{0.90\ m^2}{0.20 m^2}\right)$$

$$= 2.7 \times 10^3 N$$



2. In a car lift used in a service station, compressed air exerts a force on a small piston that has a circular cross section and a radius of 5 cm. This pressure is transmitted by a liquid to a piston that has a radius of 15.0 cm. What force must the compressed air exert to lift a car weighing 13 300 N? What air pressure produces this force?

Solution Because the pressure exerted by the compressed air is transmitted undiminished throughout the liquid, we have

$$F_1 = \left(\frac{A_1}{A_2}\right) F_2 = \frac{\pi(5.00 \times 10^{-2}\ \text{m})^2}{\pi(15.0 \times 10^{-2}\ \text{m})^2}(1.33 \times 10^4\ \text{N})$$

$$= 1.48 \times 10^3\ \text{N}$$

The air pressure that produces this force is

$$P = \frac{F_1}{A_1} = \frac{1.48 \times 10^3\ \text{N}}{\pi(5.00 \times 10^{-2}\ \text{m})^2}$$

$$= 1.88 \times 10^5\ \text{Pa}$$

Archimedes's Principle

- States that "any object, wholly or partially immersed in a fluid is buoyed up by a force equal to the weight of the fluid displaced by the object."
- the magnitude of the buoyant force is always equal to the weight of the fluid displaced by the object.

$$F_b = W_{dis.fluid}$$

$$B = (P_b - P_t)A = (\rho_{fluid} g h)A = \rho_{fluid} g V_{dis.fl}$$

Case 1: Totally Submerged Object :

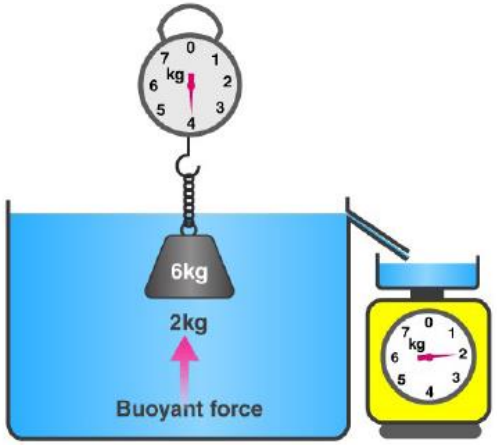
$$B = \rho_{fluid}\, g V_{dis.fl} = \rho_{fluid}\, g V_{obj},$$

F_b = buoyant force

ρ = fluid density

g = acceleration due to gravity

V = fluid volume



$$F_b = -\rho g V$$

Case 2. Partially submerged(Floating Object)

- $\rho_{obj} < \rho_{fluid}$, in static equilibrium floating on the surface of a fluid.
- In this case, the upward buoyant force is balanced by the down ward gravitational force acting on the object.

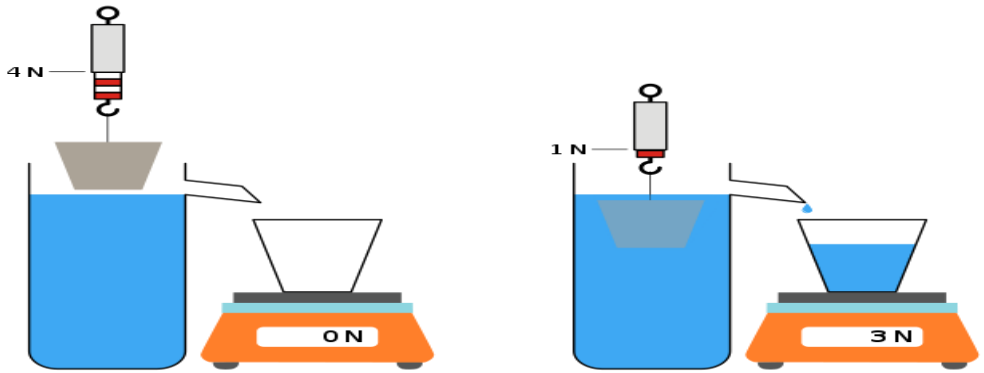
$$\frac{V_{dis.fluid}}{V_{obj(in)}} = \frac{\rho_{obj}}{\rho_{fluid}}$$

- Thus, this equation tells us that the fraction of the volume of a floating object , i.e, below the fluid surface is equal to the ratio of the density of the object to that of the fluid.

Fig. An object floating on the surface of a fluid experiences two forces, the gravitational force F g and the buoyant force B. Because the object floats in equilibrium, B = Fg .

N:B. Buoyant force can be also defined as:

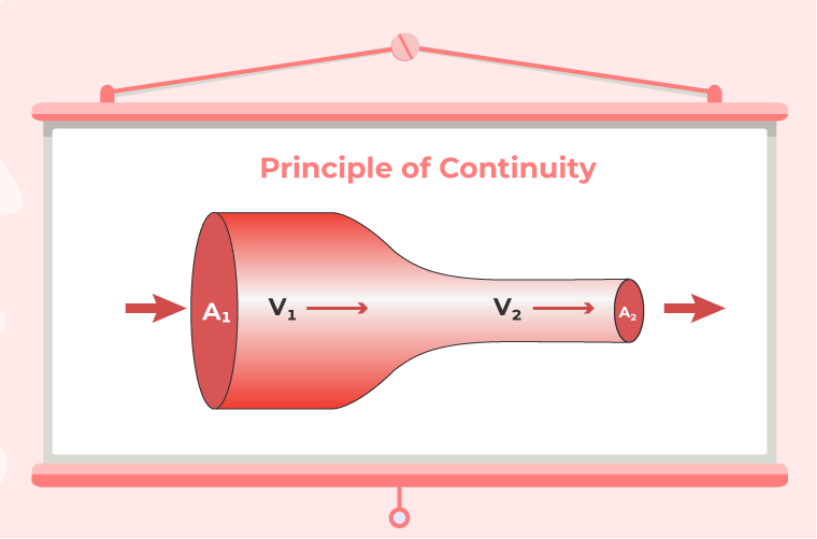
B = Weight in air- Weight in fluid



Equation of Continuity$(Conservation\ of\ mass)$

- **It states that the product of the area and the fluid speed at all points along a pipe is constant for an incompressible fluid.**

$$A_1 v_1 = A_2 v_2 = \text{constant}$$



- The equation of continuity says that the volume an incompressible fluid entering one part of a flow tube must be matched by an equal volume leaving downstream.

Note:

- An important consequence of the equation of continuity is that flow is faster in narrower parts of a flow tube, slower in wider parts.

i.e. the speed (v) is inversely proportional to Area (A)

- The product **AV** is called **volume flux or flow rate(R)** which has the dimensions of volume per unit time

. Proof of equation of continuity

In a time interval Δt, the fluid at the bottom end of the pipe moves a distance $\Delta x_1 = v_1 \Delta t$. If A_1 is the cross-sectional area in this region, then mass of fluid contained in this region is

$$m_1 = \rho A_1 \Delta x_1 = \rho A_1 v_1 \Delta t$$

Where ρ is the unchanging density of the ideal fluid

Similarly, the fluid that moves through the upper end of the pipe in the time interval Δt has a mass m_2

$$m_2 = \rho A_2 \Delta x_2 = \rho A_2 v_2 \Delta t$$

However, because the fluid is incompressible and because the flow is steady, the mass that crosses A_1 in time interval Δt must equal the mass that crosses A_2 in the same time interval.

$$mass\ flowing\ in = mass\ flowing\ out$$

$$m_1 = m_2$$

$$\rho A_1 v_1 \Delta t = \rho A_2 v_2 \Delta t$$

$$\Longrightarrow \quad \boldsymbol{A_1 v_1} = \mathrm{A}_{\mathbf{2}}\mathrm{V}_{\mathbf{2}} \quad == \text{Constant}$$

Examples

1.water flow through a pipe of circular cross-section having radius of $12cm$ at $point\ A$ and $6cm\ at\ point\ B$ as shown below. If the speed of the water at B is 340 $\frac{cm}{s}$, find is speed at A?

Solution : r_A =12cm , $A_A = \pi {r_A}^2 = \pi(12cm)^2 = 144\pi cm^2$ and $r_B = 6cm = A_B = \pi {r_B}^2 =$

$$\pi(6cm)^2 = 36\pi cm^2$$

V_A =?, $V_B = 340\frac{cm}{s}$

For a steady flow of water in pipes $A\ and\ B$

$$A_A v_A = A_B v_B \Longrightarrow 144\pi cm^2 v_A = 36\pi cm^2 x340\frac{cm}{s}$$

$$\Longrightarrow v_A = \frac{36\pi cm^2 X340\frac{cm}{s}}{144\pi cm^2} = \frac{12240\frac{cm}{s}}{144} = 85\frac{cm}{s}$$

2. If the rate of flow of water through a horizontal pipe is $280\frac{cm^3}{s}$, what is the speed of water at a point where the diameter of the pipe is $10cm$?

Solution: rate of flow $R = Av \Longrightarrow$ $v = \frac{R}{A} = \frac{280\frac{cm^3}{s}}{\pi r^2} = \frac{280\frac{cm^3}{s}}{\pi(5cm)^2} == \frac{280\frac{cm^3}{s}}{78.5cm^2} = 3.57\frac{cm}{s}$

3. The radius of **aorta** is ~ 10mm and the blood flowing through it has a speed ~ 300m.s^{-1}. A capillary has a radius ~ $4x10^{-3}mm$ but there are literally billions of them. The average speed of blood through the capillaries is ~ $5x10^{-4}m.s^{-1}$. Calculate the effective cross-sectional area of the capillaries and the approximate number of capillaries.

Solution : we are given ***aorta*** $R_A = 10x10^{-3}m$ and A_A = cross-sectional area of **aorta**

Capillaries $R_C = 4x10^{-6}m$ and A_C = cross-sectional area of capillaries

aorta $\boldsymbol{v_A}$ = $\mathbf{0.3}$0m. s^{-1}, capillaries

$v_C = 5x10^{-4}m.s^{-1}$ Assume steady flow of an ideal fluid and apply the equation of continuity

$$R = Av = constant \implies A_A v_A = A_C v_C$$

Area of capillaries A_C $A_C = A_A\left(\frac{V_A}{V_C}\right) = \pi R_A^2\left(\frac{V_A}{V_C}\right)$

$$A_C = 0.20m^2$$

Number of capillaries, N $A_C = N\pi R_C^2$

$$N = \frac{A_C}{\pi R_C^2} = \frac{0.20m^2}{\pi x(4x10^{-6})^2 m^2} \implies N = 4x10^9$$

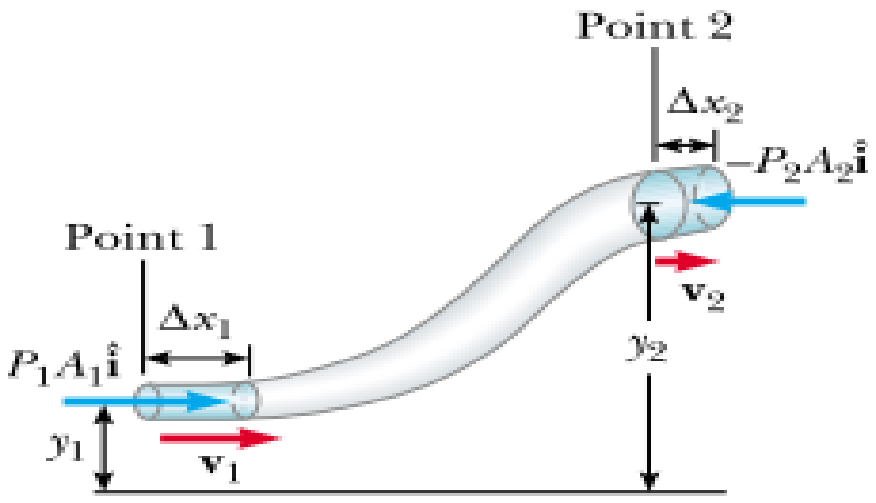
Bernoulli's principle and Equation

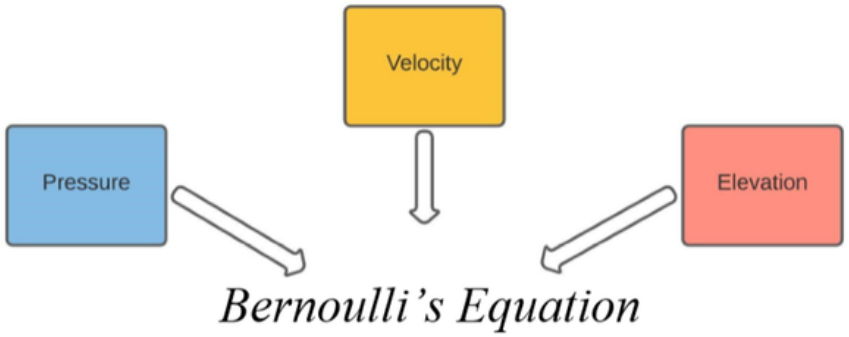
- States that "For the steady flow of a non viscous fluid through a tube, the sum of the pressure and energy of motion (kinetic energy) per unit volume and potential energy per unit volume of the fluid is constant."

$$P + \frac{1}{2}\rho v^2 + \rho g y = \text{constant}$$

- In the steady flow of a non viscous, incompressible fluid of density ρ, the pressure P, the fluid speed υ, and the elevation y at any two points (1 and 2) are related by the equation :

$$P_1 + \frac{1}{2}\rho {v_1}^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho {v_2}^2 + \rho g y_2$$



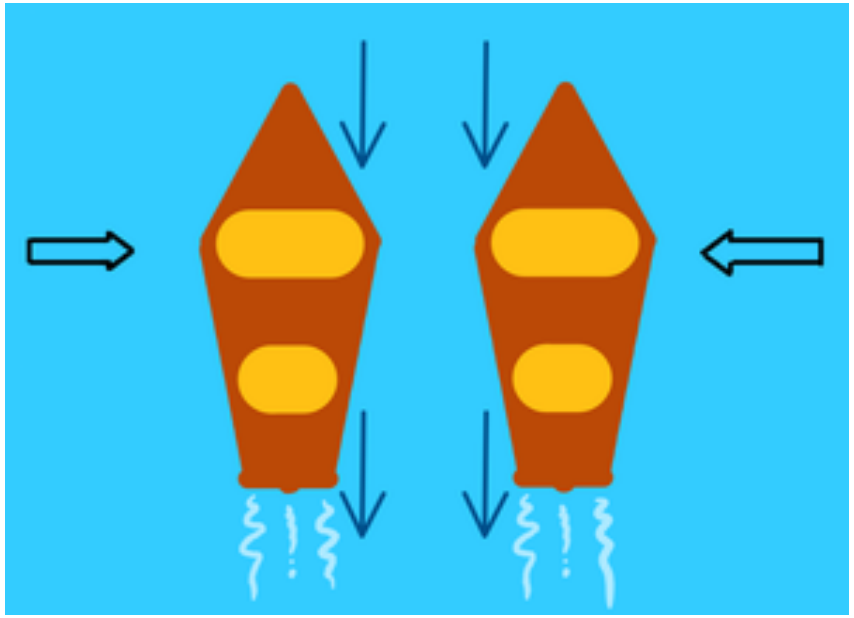


- The **Bernoulli's equation** can be derived using the **work–energy theorem.**

Applications of Bernoulli's Principle

(1) Crashing Ships

A terrifying tale of the past, two ships crashing into each other when traveling parallel to each other is a classic example of Bernoulli's Principle at play.



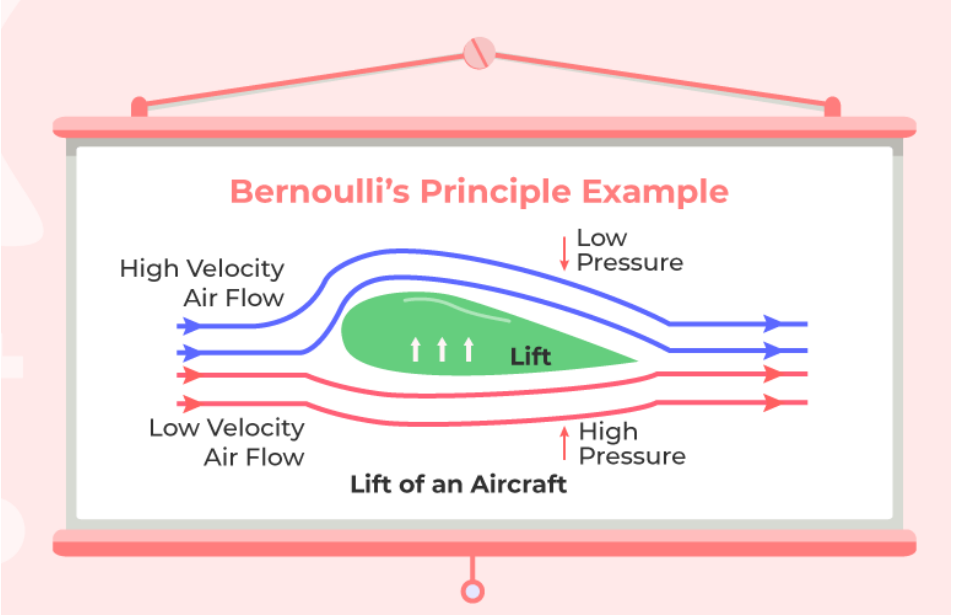
Imagine two boats sailing parallel to each other at equal velocities. Since the middle portion of the boat is the widest, it acts as a funnel or squeezed pipe for the water between the two boats. This decrease and increase in the area for the water to pass through leads to an increase in velocity (Bernoulli's Principle).

However, once the velocity increases, the pressure between the boat decreases. As a result, in order to stabilize this lower pressure the water on the outside pushes inward causing the boats to move towards each other.

2. Principle of Lifting an Aircraft

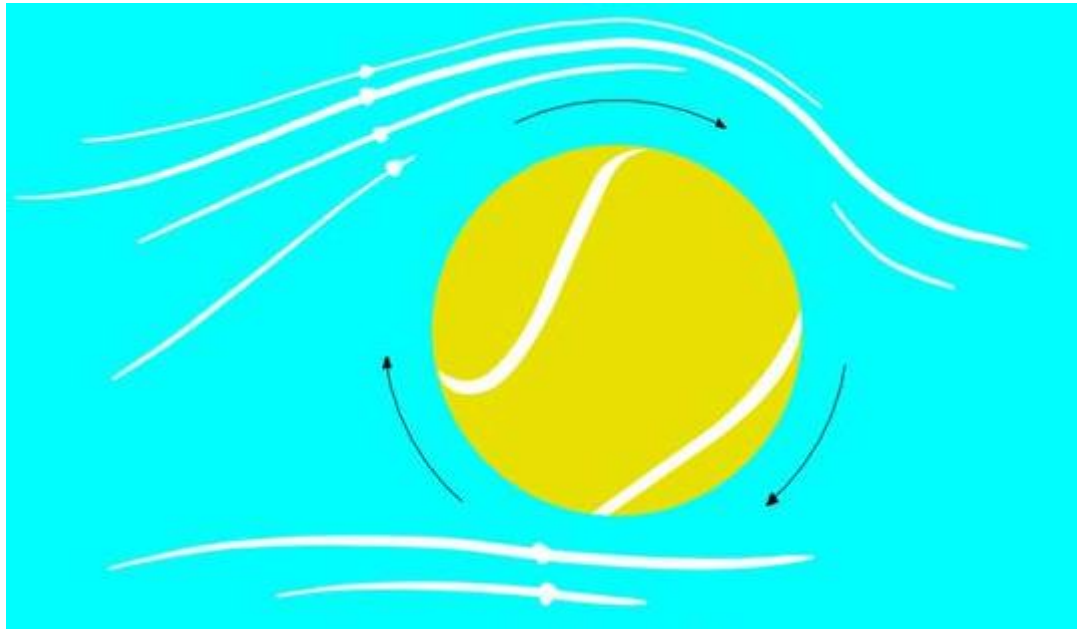
Bernoulli's theorem governs the operation of aeroplanes. The plane's wings have a certain form. When the plane is moving, the air flows past it at a high rate, despite the plane's low surface wig. There is a variation in the flow of air above and below the wings due to Bernoulli's principle.

As a result of the flow of air on the wing's up surface, this concept causes a change in pressure. If the force is greater than the plane's mass, the plane will ascend as shown in the figure below:



(3) Deviation of a Spinning Ball from its Trajectory

Any ball when thrown normally (without any spin) is observed to succumb to the effects of gravity and outline a parabolic trajectory. However, when the ball is provided a spin, it deviates from its trajectory and rises or falls depending on the direction of spin and relative air.



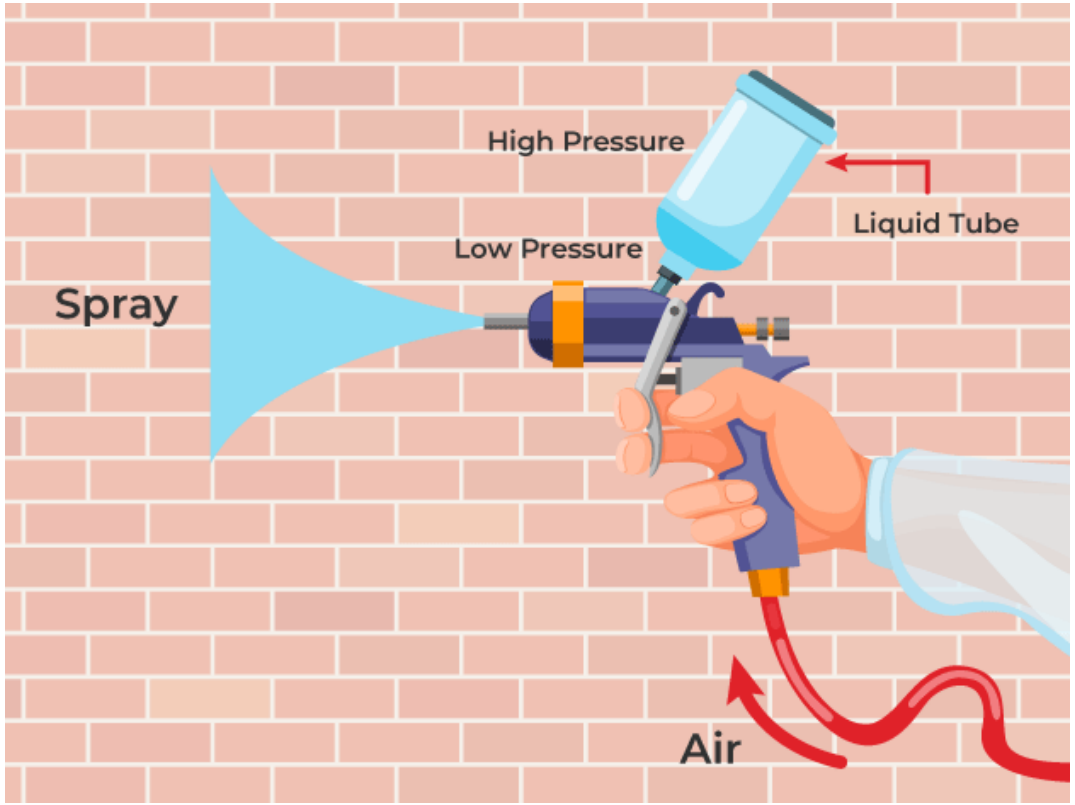
When the ball is spun and the air is moving towards the ball, the streamline of the air is dragged along with the ball, which is what we see on the top half of the figure above.

Meanwhile, the direction of air below the ball is against the spin which leads to minimal drag (the streamline continues normally). Since the air above must traverse a greater distance over and along with the ball, it must do so with greater velocity.

A higher velocity is associated with lower pressure as per Bernoulli's Principle. This difference in velocities and subsequently the generated pressures leads to a Dynamic Lift causing the ball to rise.

4. Action of Atomizer or Spray Gun

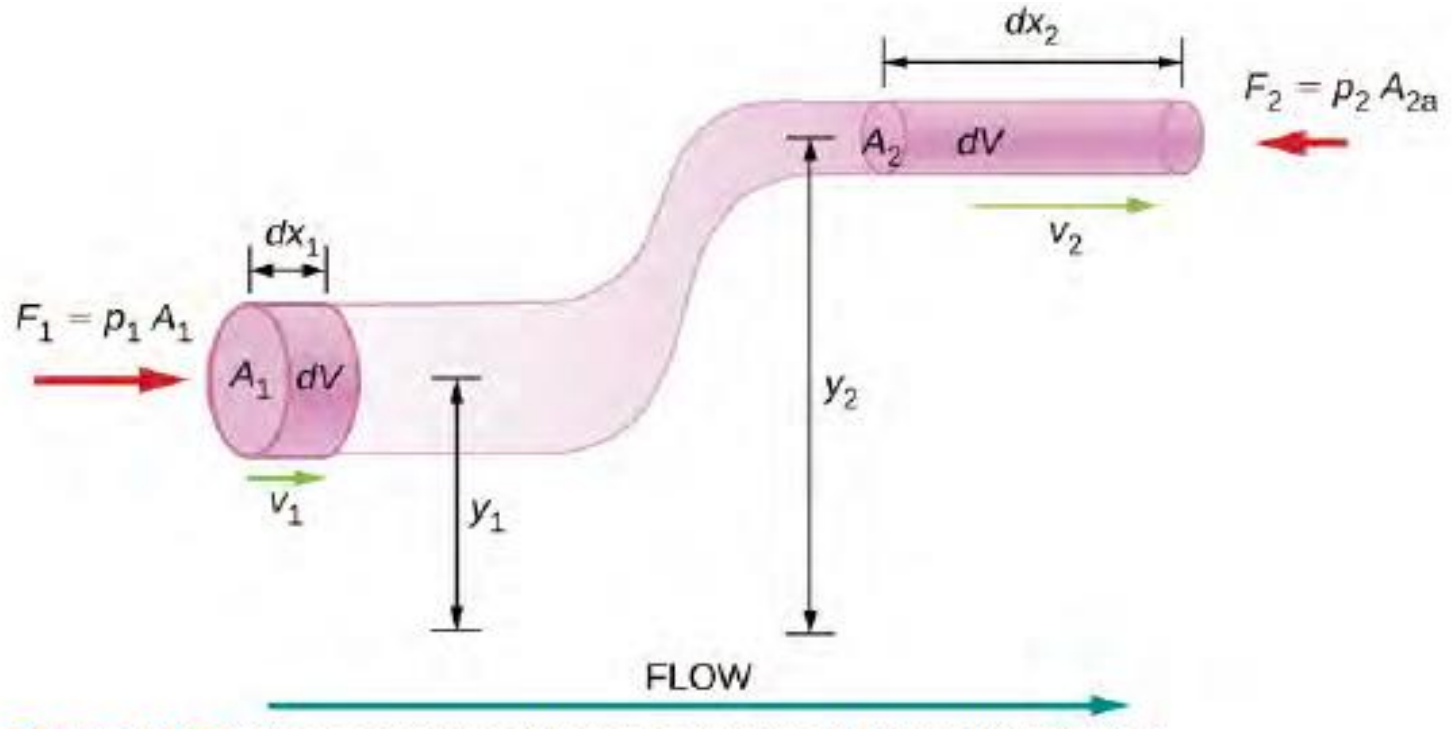
When the piston is pulled, air flows out of the horizontal tube, causing the pressure to drop which is less than the container's pressure. The consequence is that the liquid in the vertical tube rises. When it runs into the fast air in the tube, it dissipates into a thin mist. The same idea governs filter pumps, Bunsen burners, and sprayers used to apply pesticides or scents.



Proof of Bernoulli's Equation

- Consider an incompressible fluid flowing through a pipe that has a varying diameter and height, as shown in in figure below.
- We assume here that the density at the two points is the same-therefore, density is denoted by ρ without any subscripts—and since the fluid in incompressible, the shaded volumes must be equal.

- We also assume that there are no viscous forces in the fluid, so the energy of any part of the fluid will be conserved.



To derive Bernoulli's equation, we first calculate the work that was done on the fluid:

$$dW = F_1 dx_1 - F_2 dx_2$$
$$dW = p_1 A_1 dx_1 - p_2 A_2 dx_2 = p_1 dV - p_2 dV = (p_1 - p_2)dV.$$

The work done was due to the conservative force of gravity and the change in the kinetic energy of the fluid. The change in the kinetic energy of the fluid is equal to

$$dK = \frac{1}{2}m_2 v_2^2 - \frac{1}{2}m_1 v_1^2 = \frac{1}{2}\rho dV\left(v_2^2 - v_1^2\right).$$

The change in potential energy is

$$dU = mgy_2 - mgy_1 = \rho dV g(y_2 - y_1).$$

The energy equation then becomes

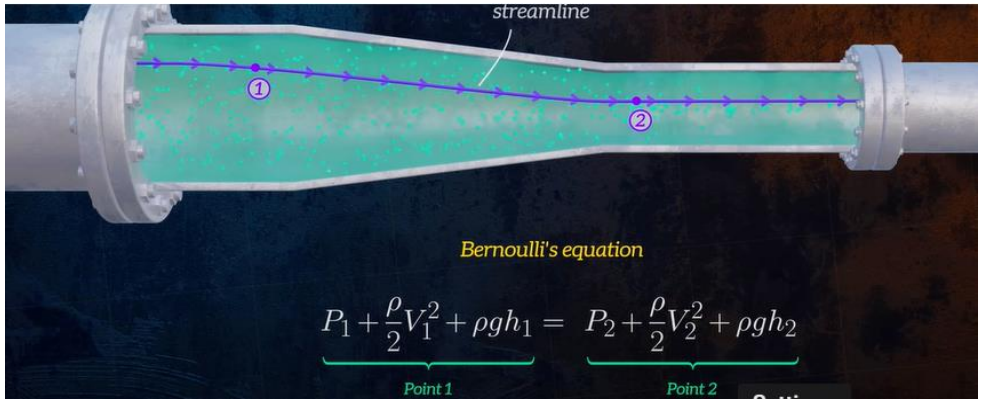
$$\begin{aligned} dW &= dK + dU \\ (p_1 - p_2)dV &= \frac{1}{2}\rho dV\left(v_2^2 - v_1^2\right) + \rho dV g(y_2 - y_1) \\ (p_1 - p_2) &= \frac{1}{2}\rho\left(v_2^2 - v_1^2\right) + \rho g(y_2 - y_1). \end{aligned}$$

Rearranging the equation gives Bernoulli's equation:

$$p_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2.$$

For a horizontal flow (H1 = H2)

There is no significant change in elevation between points 1 and 2 :



$$P_1 + \frac{\rho}{2}V_1^2 + \rho g \cancel{h_1} = P_2 + \frac{\rho}{2}V_2^2 + \rho g \cancel{h_2}$$

$$P_1 - P_2 = \frac{\rho}{2}(V_2^2 - V_1^2)$$

If we assume the fluid is incompressible, then the mass flow rate at points 1 & 2 must be equal.

- This gives us what is call us Continuity Equation, which is just a statement of conservation of mass. Mass flow rate is equal to the product of the fluid density, the pipe cross-sectional area and the fluid velocity

Continuity equation (conservation of mass)

$$\dot{m}_1 = \dot{m}_2$$

Bernoulli's equation

$$P_1 - P_2 = \frac{\rho}{2}(V_2^2 - V_1^2)$$

Continuity equation (conservation of mass)

$$\rho A_1 V_1 = \rho A_2 V_2$$

Bernoulli's equation

$$P_1 - P_2 = \frac{\rho}{2}(V_2^2 - V_1^2)$$

we can rearrange the cont. Eq to find the velocity at the 2nd point

Continuity equation (conservation of mass)

$$A_1 V_1 = A_2 V_2$$

Bernoulli's equation

$$P_1 - P_2 = \frac{\rho}{2}(V_2^2 - V_1^2)$$

$$V_2 = \frac{A_1}{A_2} V_1$$

The cross-sectional area at point 2 (A2) is less than A1. which means the velocity of the flow increases as it passes into the smaller diameter pipe. This is quite intuitive.

$$V_2 > V_1$$

$$V_2 = \frac{A_1}{A_2} V_1$$

Substituting V2 into Bernoulli's Equation, we can see since the velocity increases between points 1 and 2, the pressure b/n both points must decrease.(pressure decrease b/n point 1 and 2)

Bernoulli's equation

$$P_1 - P_2 = \frac{\rho}{2} V_1^2 \left(\frac{A_1^2}{A_2^2} - 1 \right)$$

This concept, that for a horizontal flow an increase in fluid velocity must be accompanied by a decrease in pressure , is one way of formulating Bernoulli's principle.

BERNOULLI'S PRINCIPLE

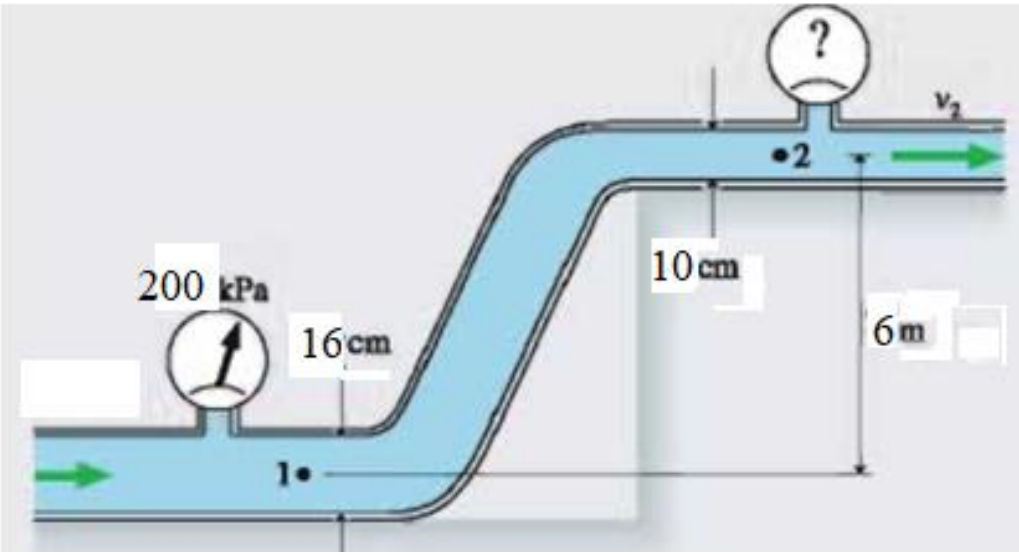
For horizontal flow an increase in velocity must be accompanied by a decrease in pressure

Examples 1. The pipe shown in fig below has a diameter of 16cm at section (1) and 10cm at section (2). At section (1) the pressure is 200kpa. Point (2) is 6m higher than point (1) when oil of density $800\frac{kg}{m^3}$ flow at a rate of $0.03\frac{m^3}{s}$. Find the pressure at point (2) if viscous effects are negligible.

Solution: from $R = V_1A_1 = V_2A_2$ we have

$$V_1 = \frac{R}{A_1} = \frac{0.03m^3.s^{-1}}{\pi(8x10^{-2}m)^2} = \frac{0.03m^3.s^{-1}}{\pi 64x10^{-4}m^2} = 1.49\frac{m}{s}$$

$$V_2 = \frac{R}{A_2} = \frac{0.03m^3.s^{-1}}{\pi(5x10^{-2}m)^2} = \frac{0.03m^3s^{-1}}{\pi 25x10^{-4}m^2} = 3.82\frac{m}{s}$$



From Bernoulli's equation :

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2 \quad \text{since } h_1 = 0$$

$$\Rightarrow \quad P_2 = P_1 + \frac{1}{2}\rho(v_1^2 - v_2^2) + \rho g(h_1 - h_2)$$

$$\Rightarrow \quad P_2 = P_1 + \frac{1}{2}\rho(v_1^2 - v_2^2) - \rho g h_2$$

$\Longrightarrow$

$$P_2 = 2x10^5 pa + \frac{1}{2}800\frac{kg}{m^3}\left[1.5\frac{m}{s} - 3.8\frac{m}{s}\right] - 800\frac{kg}{m^3}x9.8\frac{m}{s^2}x6m$$

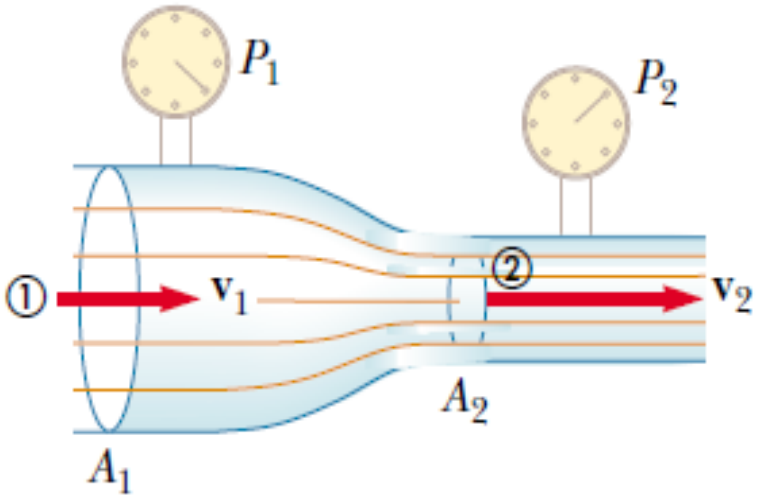
$$\Longrightarrow \quad P_2 = 2x10^5 pa + 400\frac{kg}{m^3}\left(-2.3\frac{m}{s}\right) - 47040pa$$

$$\Longrightarrow \quad P_2 = 2x10^5 pa - 4948pa - 47040pa$$

$$\Longrightarrow \quad P_2 = 2x10^5 pa - 51988pa$$

$$\therefore \quad P_2 = 1.48x10^5 pa = 148kpa$$

2. Horizontal pipe has a constriction in it as shown. At point (1) the diameter is $6cm$ while at point (2) it is only $2cm$. At point (1) $v_1 = 2\frac{m}{s}$ and $P_1 = 180kPa$. Calculate v_2 *and* P_2



Solution:

$r_1 = 3cm$ *hence* $A_1 = \pi(3cm)^2 = \pi 9cm^2 = 9\pi x10^{-4}m^2$

$r_2 = 1cm$ *hence* $A_2 = \pi(1cm)^2 = \pi cm^2 = 10^{-4}\pi m^2$

From the equation of continuity :

$$A_1 v_1 = A_2 v_2 \quad \Longrightarrow \quad v_2 = \frac{A_1 v_1}{A_2} = \frac{9x10^{-4}\pi x2ms^{-1}}{10^{-4}\pi m^2} = 18\frac{m}{s}$$

To find P_2, we use Bernoulli's equation:-

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 \;=\; P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2 \qquad \text{since } h_1 = h_2$$

$$P_1 + \frac{1}{2}\rho v_1^2 == \; P_2 + \frac{1}{2}\rho v_2^2 \qquad \Longrightarrow \quad P_2 + \frac{1}{2}\rho v_2^2 - \left[P_1 + \frac{1}{2}\rho v_1^2\right]$$

$$\Longrightarrow \quad P_2 \;=\; P_1 + \frac{1}{2}\rho(v_1^2 - v_2^2)$$

$$\Longrightarrow \quad P_2 \;= 1.8x10^5 pa + \frac{1}{2}x1000\,\frac{kg}{m^3}\left((2m.s^{-1})^2 - (18m.s^{-1})^2\right)$$

$$\Longrightarrow \quad P_2 \;= 1.8x10^5 pa + 500kg.m^{-3}[4m^2.s^{-2} - 324m^2s^{-2}]$$

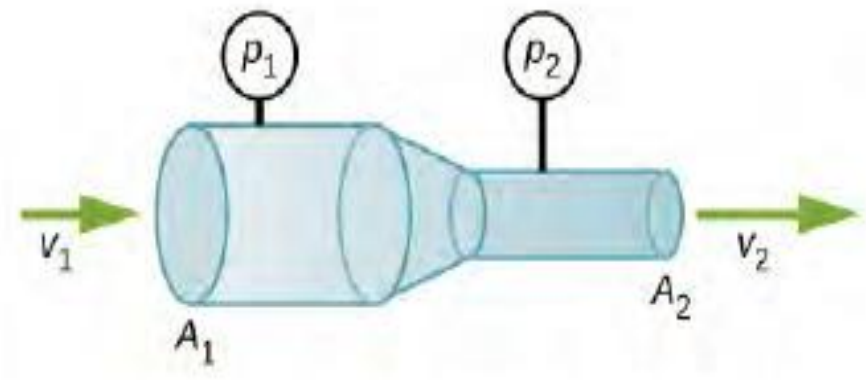
$$\Longrightarrow \quad P_2 \;= 1.8x10^5 pa + 500kg.m^{-3}(-320m^2.s^{-2})$$

$$\Longrightarrow \quad P_2 \;= 1.8x10^5 pa - 160000pa$$

$$\Longrightarrow \quad P_2 \;= 1.8x10^5 pa - 1.6x10^5 pa \qquad = 0.2x10^5 pa$$

$$\therefore \qquad P_2 \;= 20kpa$$

3. In the horizontal constricted pipe illustrated in figure below. Determine the flow speed at point 2 if the pressure difference P1 & P2 is known.



Solution Because the pipe is horizontal, $y_1 = y_2$, and applying Equation 14.8 to points 1 and 2 gives

$$(1) \qquad P_1 + \tfrac{1}{2}\rho v_1^2 = P_2 + \tfrac{1}{2}\rho v_2^2$$

From the equation of continuity, $A_1 v_1 = A_2 v_2$, we find that

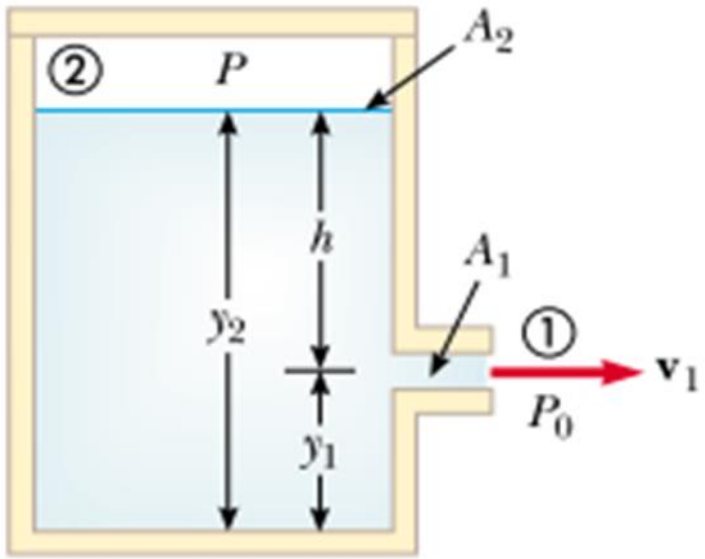
$$(2) \qquad v_1 = \frac{A_2}{A_1} v_2$$

Substituting this expression into Equation (1) gives

$$P_1 + \tfrac{1}{2}\rho\left(\frac{A_2}{A_1}\right)^2 v_2^2 = P_2 + \tfrac{1}{2}\rho v_2^2$$

$$v_2 = A_1\sqrt{\frac{2(P_1 - P_2)}{\rho(A_1^2 - A_2^2)}}$$

4. An enclosed tank containing a liquid of density ρ has a hole in its side at a distance y_1 from the tank's bottom as shown in fig below. The hole is open to the atmosphere, and its diameter is much smaller than the diameter of the tank. The air above the liquid is maintained at a pressure P. Determine the speed of the liquid as it leaves the hole when the liquid's level is a distance h above the hole.



Solution

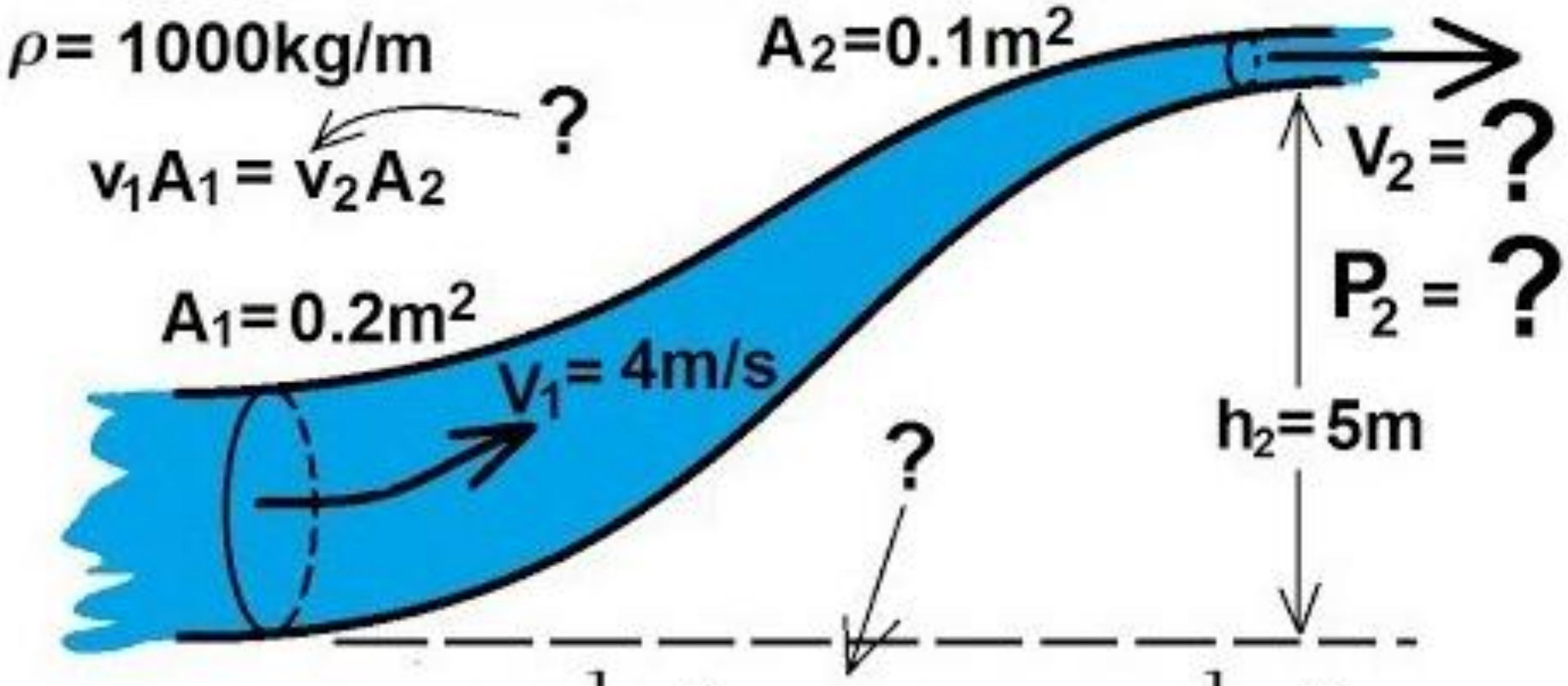
Because A2 >> A1,(i.e V2 << V1), the liquid is approximately at rest at the top of the tank, where the pressure is P. Applying Bernoulli's equation to points 1 and 2 and noting that at the hole P1 is equal to atmospheric pressure P0, we find that

$$P_0 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P + \rho g y_2$$

But $y_2 - y_1 = h$; thus, this expression reduces to

$$v_1 = \sqrt{\frac{2(P - P_0)}{\rho} + 2gh}$$

Bernoulli's Eqn: Unknown Velocity

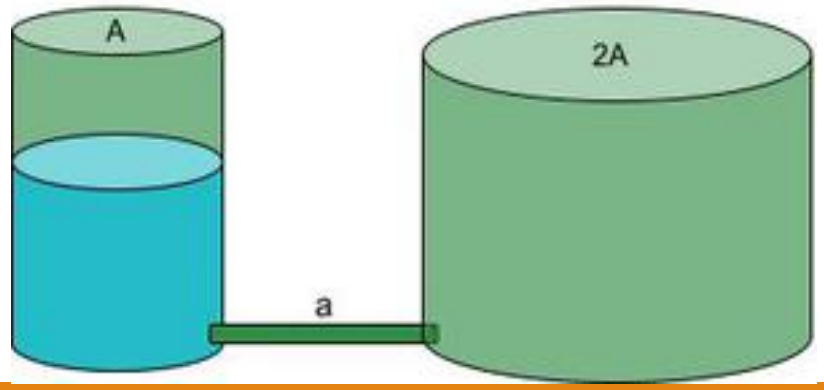


$$P_1 + \rho g h_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g h_2 + \frac{1}{2}\rho v_2^2$$

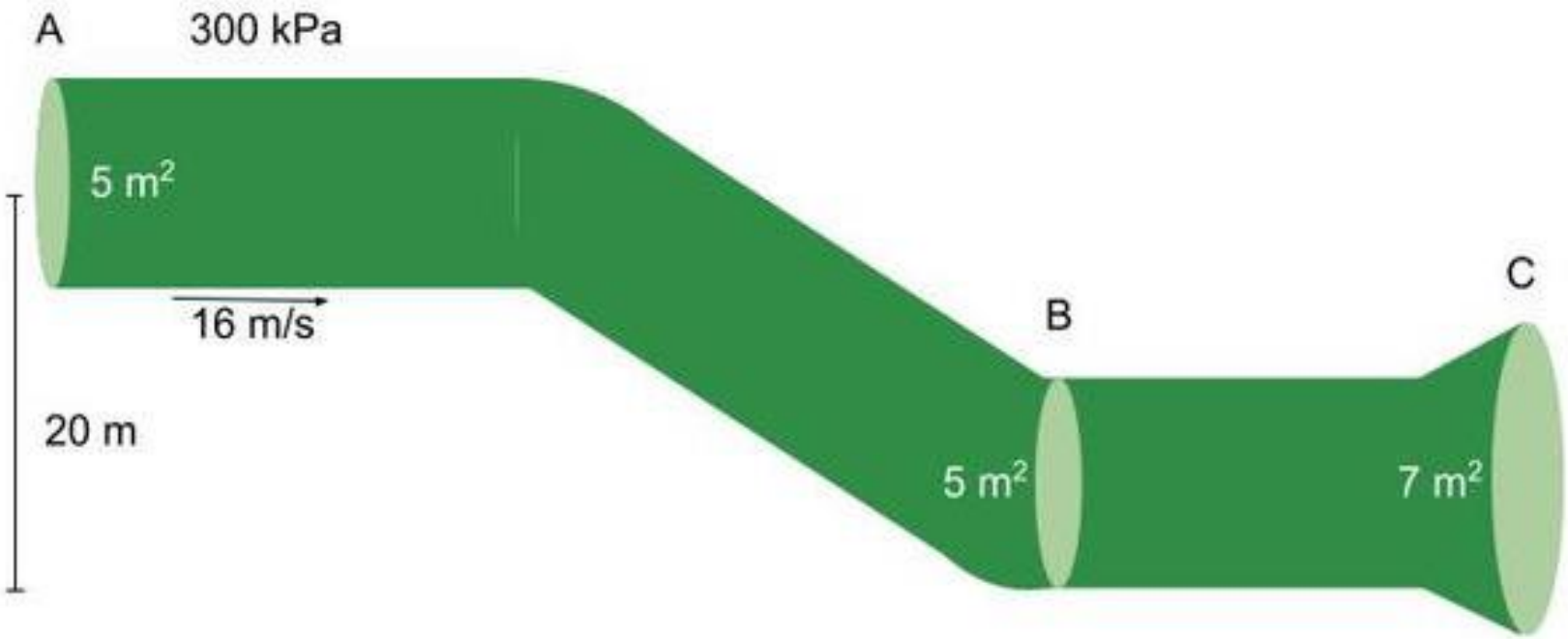
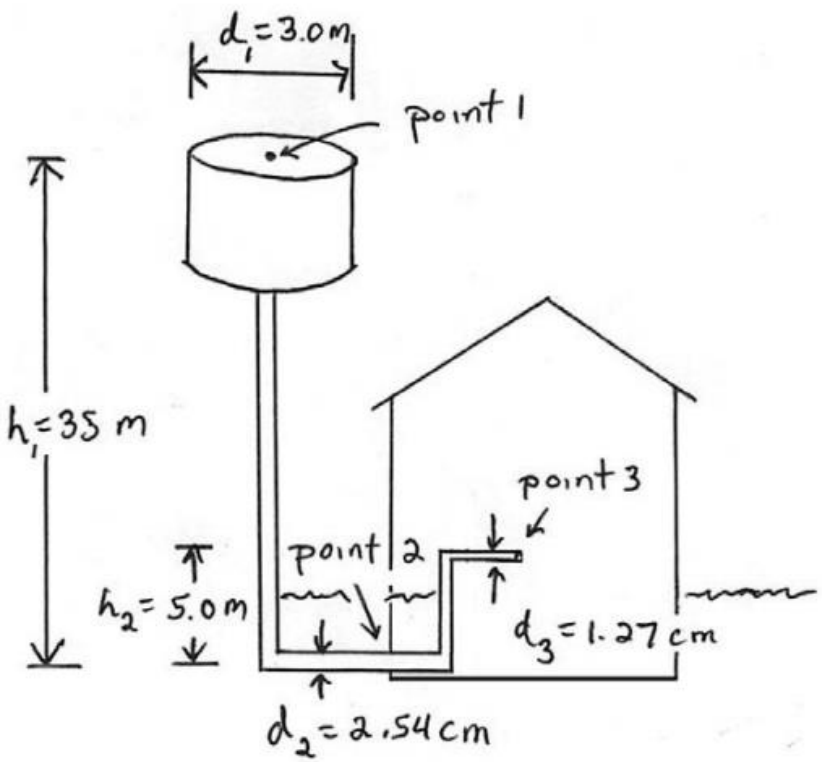
Ass #2

1. A cylindrical water tower of diameter 2 m supplies water to a house. The level of water in the water tower is 50 m above the point where the water enters the house through a pipe that has an inside diameter 4cm. The intake pipe delivers water at a maximum rate of 3×10^{-3} $m^3.s^{-1}$. The pipe is connected to a narrower pipe leading to the second floor that has an inside diameter 3.5 cm. Calculate the pressure of the water in the narrow pipe at 4 m height. Find the speed of the water in the narrow pipe. (Use: Patm = 1atm = 101.3 KPa, ρ = 1000 Kg/m^3 , g = 10 m/s^2

2. Two cylindrical tanks having cross-sectional areas A and 2A respectively are kept on a horizontal floor. The first tank is filled to a height of H while the other is empty. If they are connected by a pipe with negligible cross-sectional are 'a' at the bottom at time t=0, find how long it would take for the heights in the two tanks to become equal.



3. Refer to the figure below and arrange the pressures at points A, B, and C in ascending order. Do the same for the speeds at points A, B, and C.



4. A cylindrical water tower of diameter 3.0 m supplies water to a house. The level of water in the water tower is 35 m above the point where the water enters the house through a pipe that has an inside diameter 5.1 cm. The intake pipe delivers water at a maximum rate of $2.0\times10^{-3}\,m^3\cdot s^{-1}$. The pipe is connected to a narrower pipe leading to the second floor that has an inside diameter 2.5 cm . What is the pressure and speed of the water in the narrower pipe at a point that is a height 5.0 m above the level where the pipe enters the house?